NeuroEngineering

Lecture Notes

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Preface & Acknowledgments

❖ Lecture 1

1.1 Historical Background

The **Electroencephalogram** (EEG) is the non-invasive detection of electrical signals from the scalp generated by cortical neurons.

The alpha rhythm is a spontaneous EEG oscillation typically ranging between 8 and 13 Hz, first identified by Hans Berger in the 1920s. Historically, based on the observations of Lord Adrian and Matthews, it was considered an "idling" state of the brain. Because alpha synchronization is most prominent over the occipital cortex during relaxed, eyes-closed states and desynchronizes upon opening the eyes or engaging in visual tasks, it was originally thought to merely represent functional inactivity.

However, modern research challenges the strict idling hypothesis, revealing that alpha oscillations also play an active role in cognitive processing. Studies have shown that alpha rhythms can emerge even with open eyes during visual working memory tasks.

1.2 The brain

The brain can be modeled as a complex, non-linear, time-variant dynamical system, where the waves recorded on the scalp represent the macroscopic output of neural activity. Because these spontaneous oscillations vary between individual trials even for the same stimulus, it is standard practice to average multiple recordings to extract consistent, time-locked signals, such as **Event-Related Potentials** (ERPs). An ERP is a change of electrical potential registered immediately after a given input to the brain, that are also spontaneous rythms that do not need a time defined event. The averaging procedure is a key concept to be capable of distinguish between the ERPs and the electrical noise (called "Artifact").

To understand the amplitude of these macroscopic EEG waves, it is crucial to introduce the concept of **Synchronization**. A brain rhythm is a neurophysiological phenomenon that directly reflects the coordinated firing of neuronal populations. The amplitude of the EEG signal is strictly tied to this coordination:

neural synchronization results in an amplitude increase, whereas desynchronization leads to an amplitude decrease.

A classic example is the alpha rhythm, which synchronizes (increases in amplitude) when a subject is resting with eyes closed, and desynchronizes (decreases in amplitude) upon opening the eyes.

This principle also extends to ERPs: early, low-latency components reflect highly synchronized activity within primary cortices, whereas later components involve more desynchronized, widely distributed neural networks.

1.3 Technical Specifications and Strumentation

The EEG is measured through an electrode, which is a very sensible voltmeter that captures changes in the electrical potential on the surface where it is placed. It is crucial to understand the sensibility and the precision of this tool. Indeed, the order of the signal is micro-Volts, around typically 5–50 µV, rarely up to 100 µV.

To obtain a measurement of the signal over time, more than one electrode are placed on the surface.

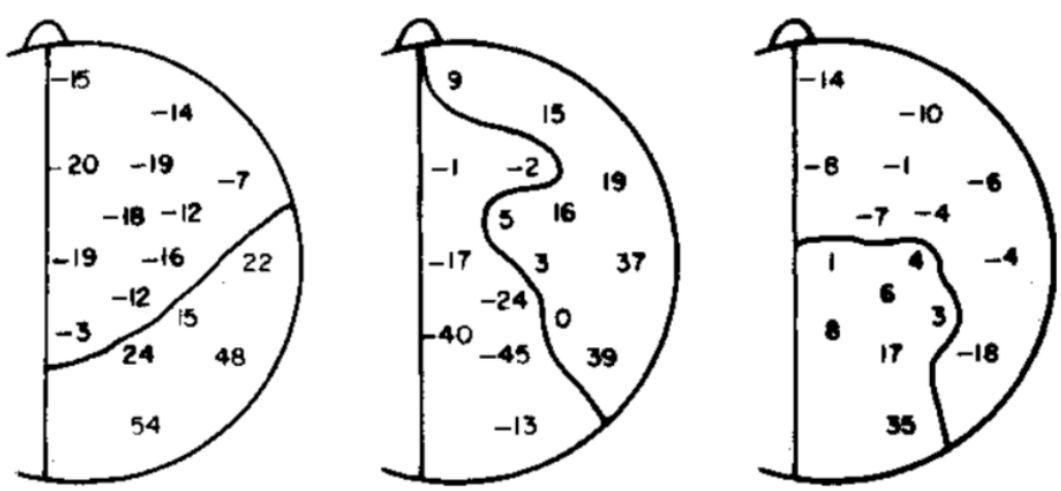
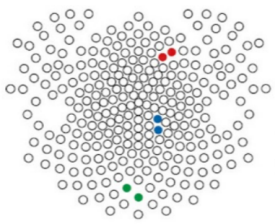
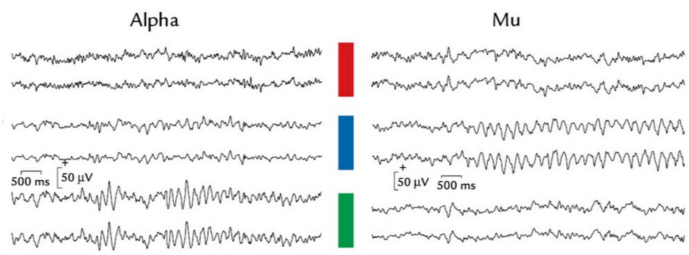


Figure 1: Map of the distribution of potential on the scalp at different times.

By placing the electrods we can study the main part of the brain which is responsible for a determined rithym. In the following picture we can analyse the Alpha and Mu rithms, which many electrodes placed on the scalp surface.



(a) Positioning of the electrodes on a flattened scalp view (Red: frontal; Blue: motor cortex; Green: occipital).

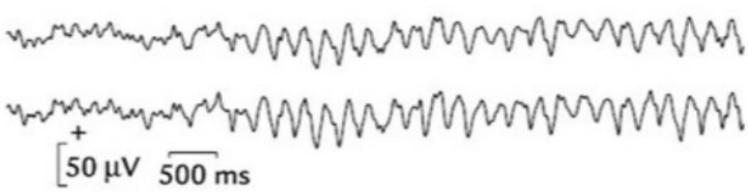


(b) Analysis of Alpha and Mu rhythms among different electrodes' placements.

Figure 2: Rythm response depending on the electrodes' positions.

We can notice, as we were explaining before, that the alpha rhythym is highly influenced by the eyes activity, which is expressed by the occipital region of the brain. In the same way, we can notice that the motor cortex is the brain region which describes more the mu rhythm.

The bars on the graphs are the so called *calibration bars*, needed to show the scale of the timeline and the amplitudes of the signals.



By this bar we can notice the ERP ares in the order of micro-Volts [μV], and there also is a positive [+] or negative [-] sign to better understand the sign of the oscillation with respect to the time (represented on the orizzontal bar). This process is called **Calibration**.

1.4 Brain Rhythms

Brain activity is characterized by oscillations at various frequencies, which are grouped into specific bands. Each rhythm is associated with particular brain regions and physiological or cognitive states.

- **Alpha Rhythm (8–13 Hz):** Traditionally considered an "idling" rhythm, it is most prominent over the occipital lobe (visual cortex). Its amplitude is highly responsive to visual input: it increases (synchronizes) when the eyes are closed and decreases (desynchronizes) when the eyes are open. The amplitude typically waxes and wanes over seconds.

- **Mu Rhythm (8–13 Hz):** This is the motor-cortex counterpart of the alpha rhythm. It is located over the central (sensorimotor) cortex and is primarily visible on central electrodes (often the blue channels in topographic maps). It frequently exhibits a "comb-shaped" waveform with asymmetric peaks.

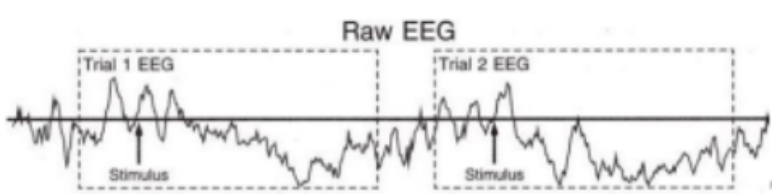
- **Beta Rhythm (14–30 Hz):** Associated with active motor and cognitive processing. It is observed across multiple cortical areas. The typical band is 15–30 Hz, though it is often cited around 20–25 Hz.

- **Theta Rhythm (4–7.5 Hz):** Linked to memory, workload, and stress. It is prominent in midline structures, such as the hippocampus and frontal midline, and its amplitude increases during demanding cognitive tasks.

- **Delta Rhythm (< 3.5 Hz):** Characterized by very slow waves (typically < 3.3 Hz). In healthy adults, these are rare during wakefulness and usually indicate deep sleep or pathological states.

- **Gamma Rhythm (> 30 Hz):** They are correlated with large-scale brain network activity and cognitive phenomena such as working memory, attention. Altered gamma activity has been observed in many mood and cognitive disorders such as Alzheimer's disease, epilepsy, and schizophrenia.

An **Event Related Response** (ERP) is a potential related to a event (cognitive input) of the brain.



They are also shown by the EEG, as a response to various sensory stimuli. Their latency after the stimulus can range from a few milliseconds to several hundreds of milliseconds. Usually, the fastest responses come from the *primal areas* of the brain, the earlier in the stage of processing. A **trial** is a segment of the EEG after

a sensorial stimulus, and the **latency** is the time between the stimulus and the potential/trial. To analyse a global response, many individual trials are taken and averaged.

References